

# Threat Model

**Grounded in:** Insulet Omnipod 5 (AID patch pump)

**Basis:** STRIDE; device facts public.

|              |                                                                                      |
|--------------|--------------------------------------------------------------------------------------|
| Device       | Insulet Omnipod 5 — automated insulin delivery (AID) patch pump                      |
| Connectivity | Bluetooth to compatible smartphone / Controller; CGM input drives dosing every 5 min |
| Method       | STRIDE over pod, CGM link, phone/controller, cloud                                   |
| Note         | Illustrative; safety-critical dosing.                                                |

## Threats & mitigations (STRIDE)

| Threat    | Example                          | Mitigation                                     | V&V                |
|-----------|----------------------------------|------------------------------------------------|--------------------|
| Spoofing  | Rogue controller/CGM peer        | Bonded pairing, authenticated link             | BLE security tests |
| Tampering | Altered dose command / CGM value | Signed commands, dose limits, value validation | Fuzz + limit tests |
| DoS       | Loss of CGM/controller link      | Safe fallback to basal; alerts                 | Fault injection    |
| Elevation | Cloud/app abuse                  | Least-privilege, scoped tokens                 | IAM review         |

## Residual risk

Maximum-dose limits and safe-state basal bound worst-case harm if a link is degraded; disclosed in labeling.

## Sources (public)

- Insulet — Omnipod 5, first FDA-cleared tubeless automated insulin delivery (AID) patch pump; smartphone-controlled.
- Public facts: integrated dosing algorithm adjusts insulin every 5 min using CGM input; 510(k) algorithm update (Dec 2025).
- FDA — Premarket Cybersecurity guidance (2025); PCCP final guidance (2024); GMLP.